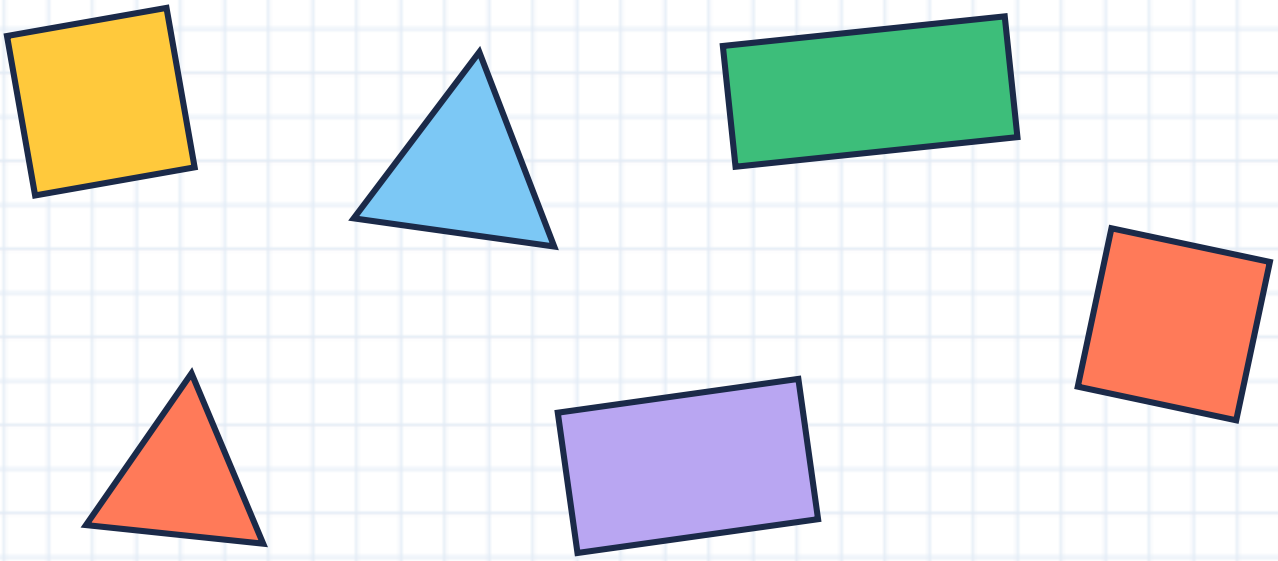


LEVEL 4 · HOLIDAY PROGRAM

Math Inquirers

Notice it. Test it. Explain why.

Name _____ **Group** _____

My Inquirer Rules

1. Notice the pattern.
2. Test it with objects first.
3. Check your answer.
4. Explain why.

My Inquiry Journal

Your tutor stamps each lesson.

Lesson 1
Factor Hunters

Lesson 2
Big Number Puzzles

Lesson 3
Operation Detectives

Lesson 4
Fraction Frontier

Lesson 5
Decimal Detectives

Lesson 6
Angle Turns

Lesson 7
Area Architects II

Lesson 8
Symmetry and Nets

Lesson 9
Data Detectives

Lesson 10
Challenge Day

Tutor: _____ Program dates: _____

FACTOR HUNTERS

Name _____ Group _____ Date _____

Page 1: Start here. *After question 3 you may choose: carry on, or try Think harder.*

No.1

Hunt down every factor of 48. (A factor goes into 48 exactly.)

a Write the factor pairs of 48 in order. You may not need every box.

1 × 48	__ × __	__ × __	__ × __	__ × __	__ × __

b List all the factors of 48 in order.

Factors: _____

c How many factors does 48 have? _____ How did you know when to stop looking?

.....

No.2

A gardener plants 40 trees in equal rows. There are at least 2 rows, and at least 2 trees in each row. (2 rows of 20 trees and 20 rows of 2 trees are different ways to plant.)

a Find every way. One way is 4 rows of 10 trees. Fill in the table. You may not need every column.

Rows	4						
Trees in each row	10						

How many ways? _____

No.3

Mei has 18 red beads and 24 blue beads. She puts them into identical packs. Every pack has the same number of red beads and the same number of blue beads. No beads are left over.

a Factors of 18: _____ Factors of 24: _____

b Common factors of 18 and 24: _____

c Greatest number of packs: _____ Each pack has _____ red and _____ blue.

Page 2 is on your own. Aim for Q4 and Q5 first. Q6 is a choice for the last few minutes.

No.4

At an interchange, Bus 33 leaves every 10 minutes and Bus 51 leaves every 15 minutes. Both buses leave at 6:00 am.



a Bus 33: 6:00, 6:10, _____, _____ … Bus 51: 6:00, _____, _____ …

They next leave together at: _____

b How many times do they leave together from 6:00 am to 8:00 am? Count 6:00 and 8:00 too.

Number of times: _____

No.5

Count the factors.

a Fill in the table.

Number	6	9	10	15	16	25
How many factors?						

b Which of these numbers have exactly 3 factors? What do you notice about them?

c Is 49 a number with exactly 3 factors? List its factors to check. How could your answer to (b) have told you?

Factors of 49: _____

No.6

Multiple riddles.

a I am a number between 30 and 70. I am a multiple of 3 and a multiple of 4, but I am NOT a multiple of 8. My ones digit is bigger than my tens digit. Who am I?

Step 1: multiples of 3 AND 4 between 30 and 70 (count in 12s): _____, _____, _____. Step 2: cross out any multiple of 8. Left: _____ and _____. Step 3: use the last clue. I am: _____

b Ravi says: "Every multiple of 6 is also a multiple of 3." Mei says: "Every multiple of 3 is also a multiple of 6." Who is right? Explain with an example.

Investigation and Looking back

Investigation

Bus meeting times. ***What if we change …? What stays the same? Find all the ways.*** **Bus A leaves a stop every 6 minutes. Bus B leaves every 8 minutes. Both leave at 7:00 am and next leave together after 24 minutes. Now keep Bus A at 6 minutes and change the gap for Bus B.**

> **Easy first step.** Write the first five times Bus A leaves (7:00, 7:06, 7:12, …) and the first five for Bus B (7:00, 7:08, …). Circle a time that is in both lists. Then carry on with the parts below, as far as you like.

a Bus B every 9 minutes: they next leave together after _____ minutes. Every 12: _____. Every 5: _____. Every 7: _____.

b What stays the same every time? Write what you notice about the meeting time.

c Find ALL the pairs of different gaps (each from 2 to 12 minutes) for two buses that next leave together after exactly 12 minutes. One pair is 3 and 4.

Looking back

Looking back 2 minutes at the end. You can write, tick or pass. Nobody marks this. It is not shown to the group. Your tutor may look at it on camera if you hold it up.

Something I did well _____

The hardest part was _____

What helped me ☐ drawing it ☐ a list or table ☐ a smaller number ☐ my partner ☐ reading it again ☐ I asked for help ☐ other: _____

☐ **I am still stuck** (my tutor will check in with me) ☐ **I pass today** (that is fine)

Story corner: Sophie Germain France (Paris), 1776–1831

Sophie Germain taught herself maths from her father's books in Paris, over 200 years ago. She loved prime numbers, the numbers with exactly two factors.

Try it (optional): Is 2 × 7 + 1 a prime? List the factors of your answer. Then try 5.

Think harder

Draw a model, make a list or make a table. Then write how you know. Choose these any time, or take them home.

No.1

How many factors does 72 have? List them in order.

No.2

One lighthouse flashes every 6 seconds. Another flashes every 9 seconds. They flash together now. After how many seconds will they next flash together?

No.3

Find all the common factors of 24 and 36.

No.4

Find ALL the numbers from 1 to 60 that have exactly 3 factors. What do the numbers have in common?

Go further

No.5

Bus A leaves a stop every 6 minutes and Bus B every 9 minutes. Bus C leaves every ____ minutes. All three leave together at 9:00 am, and they next leave together at 9:36 am. What could the gap for Bus C be? Find ALL the whole-minute answers. Then make up a puzzle like this one for a partner.

No.6

I am a number less than 100. When I am divided by 2, 3, 4, 5 or 6, there is always 1 left over. I am bigger than 1. Who am I?

No.7

There are 20 closed lockers, numbered 1 to 20. Child 1 opens every locker. Child 2 changes every 2nd locker (opens it if it is closed, closes it if it is open). Child 3 changes every 3rd locker, and so on, up to child 20. Which lockers are open at the end?

BIG NUMBER PUZZLES

Name _____ Group _____ Date _____

Page 1: Start here. *After question 3 you may choose: carry on, or try Think harder.*

No.1

Round each number. Draw a number line if it helps.

Number	to the nearest 10	to the nearest 100	to the nearest 1000
47 285			
63 950			

Try next, if you have time: 9049 → _____ _____ _____ 80 503 → _____ _____ _____

No.2

Work backwards.

a The number of people at a concert is 63 000 when rounded to the nearest thousand. What is the smallest and the largest number of people there could be?



Start here: 63 500 is halfway between 63 000 and 64 000. Does it round up or down? _____ Smallest: _____ Largest: _____

b A whole number rounds to 5350 to the nearest 10. Find ALL the numbers it could be. Start at 5340 and check each one.

Numbers: _____

No.3

A bakery bakes 4982 buns every day for 6 days.

a Round 4982 to the nearest thousand and estimate the total. Estimate: _____

b Is the exact total more or less than your estimate? How do you know without working it out?

c Work out the exact total. Exact total: _____

Page 2 is on your own. Aim for Q4a, Q5 and Q6a first. The other parts are a choice for the last few minutes.

No.4

Number riddles. Who am I?

a I am a 5-digit number. Rounded to the nearest thousand, I am 48 000. My hundreds digit is 6. My ones digit is 1 and my tens digit is double my ones digit.

Hint: the numbers that round to 48 000 are from 47 500 to 48 499. Which of them have 6 in the hundreds place?

I am: _____

b I am a 5-digit number. My digits add up to 2. Find ALL the numbers I could be. List them in order.

.....

.....

How many numbers? _____

No.5

Find the rule. Write the next numbers.

a 23 450, 24 450, 25 450, _____, _____

b 81 300, 79 800, 78 300, _____, _____

c 99 950, 99 960, 99 970, 99 980, _____, _____

d 12 500, 15 000, 17 500, 20 000 ... What is the first number in this pattern that is more than 30 000? _____

No.6

Use the digit cards 0, 3, 5, 6 and 8 once each to make a 5-digit number.



a The largest even number: _____

b The smallest odd number: _____

c The number closest to 60 000: _____

Investigation and Looking back

Investigation

Double rounding. ***Is it always true? Test and explain.*** **Mei says: "If I round a number to the nearest hundred, and then round that answer to the nearest thousand, I always get the same answer as rounding to the nearest thousand straight away."**

> **Easy first step.** Round 3 280 to the nearest hundred. Then round your answer to the nearest thousand. Now round 3 280 straight to the nearest thousand. Did you get the same answer both ways? Then carry on with the parts below, as far as you like.

a Test Mei's idea on at least 5 numbers. Show each test on the lines (number, nearest hundred, then nearest thousand, and straight to the nearest thousand). Include 3 280, 1 449 and 1 450.

b Is Mei always right? If not, write one number that breaks her idea.

c Look at the numbers that break it. What do their last three digits have in common?

.....

.....

.....

.....

.....

Looking back 2 minutes at the end. You can write, tick or pass. Nobody marks this. It is not shown to the group. Your tutor may look at it on camera if you hold it up.

Something I did well _____

The hardest part was _____

What helped me ☐ drawing it ☐ a list or table ☐ a smaller number ☐ my partner ☐ reading it again ☐ I asked for help ☐ other: _____

☐ **I am still stuck** (my tutor will check in with me) ☐ **I pass today** (that is fine)

Story corner: Leonardo of Pisa (Fibonacci) Italy (Pisa), schooled in Bugia, now Bejaia, Algeria, about 1170 - about 1240/1250

Leonardo of Pisa (Fibonacci) learned the digits 0 to 9 as a boy in North Africa. His book, written in 1202, helped Europe use place value.

Try it (optional): Use the digits 7, 0, 4, 9 and 2 once each. Write the biggest and the smallest 5-digit number you can.

Think harder

Draw a model, make a list or make a table. Then write how you know. Choose these any time, or take them home.

No.1

I am a 5-digit number. All my digits are the same. Rounded to the nearest ten thousand, I am 60 000. Who am I?

No.2

Use the digit cards 1, 4, 6, 7 and 9 once each. Make the 5-digit number that is closest to 70 000.

No.3

Find the rule and write the next two numbers: 17 250, 19 500, 21 750, _____, _____

No.4

How many whole numbers round to 3500 when rounded to the nearest hundred?

Go further

No.5

How many 5-digit numbers have digits that add up to 3?

No.6

Make up your own number riddle about a 5-digit number. Use a rounding clue, a clue about the digits and an odd-or-even clue, so that exactly ONE number fits. Check your riddle: list every number that fits the rounding clue, then cross out the ones that break the other clues. Swap riddles with a partner.

No.7

The distance counter on Mr Tan's car shows 39 993 km. This number is a palindrome: it reads the same forwards and backwards. What is the next palindrome the counter will show? How many more kilometres is that?

OPERATION DETECTIVES

Name _____ Group _____ Date _____

Page 1: Start here. *After question 3 you may choose: carry on, or try Think harder.*

Brackets first. Then × and ÷, working from left to right. Then + and –, working from left to right.

No.1

Work out. A story helps: 36 + 14 × 5 means 36 stickers and then 5 more packs of 14 stickers. Underline the part you do first.

a 36 + 14 × 5 = _____

b (36 + 14) × 5 = _____

c 72 ÷ 8 × 3 = _____

d 100 – 45 ÷ 5 + 7 = _____

e 6 × (25 – 17) ÷ 4 = _____

No.2

Missing signs. Write +, –, × or ÷ in each circle to make the number sentence true. You may use a sign more than once.

(a) 7 ◯ 6 ◯ 4 = 38

(b) 48 ◯ 6 ◯ 2 = 16

(c) 9 ◯ 3 ◯ 5 = 8

(d) 12 ◯ 3 ◯ 2 ◯ 5 = 14

No.3

Brackets change the answer. Look at 4 + 8 × 5 – 3

a With no brackets: 4 + 8 × 5 – 3 = _____

b Put in ONE pair of brackets to make 57: 4 + 8 × 5 – 3 = 57

c Put in ONE pair of brackets to make 20: 4 + 8 × 5 – 3 = 20

Page 2 is on your own. Aim for Q4, Q5 and Q6 first. Q7 is a choice for the last few minutes.

No.4

A number goes into this machine chain. 30 comes out.



a Work backwards. What number went in? _____ **b** Check it going forwards.

No.5

Number trick. Think of a number. Multiply it by 5. Add 35. Divide by 5. Take away the number you started with. Try it with two different numbers.

My number	× 5	+ 35	÷ 5	– my number

a What do you notice? _____

b Fill in the blanks. × 5 then ÷ 5 gives back my _____. The 35 ÷ 5 adds _____. Taking away my number leaves _____.

No.6

A hawker stall sold 3 times as many plates of chicken rice as plates of laksa. It sold 124 plates altogether.

a Draw a bar model.

b How many plates of chicken rice did it sell? _____

c How many more plates of chicken rice than laksa did it sell? _____

No.7

Number chains. After the first two numbers, each number is the sum of the two numbers before it. Example: 2, 5, 7, 12, 19. Fill in the missing numbers.

(a) ☐, ☐, 13, 21, ☐

(b) 4, ☐, ☐, 18, ☐

Hint: two neighbours known? Work backwards (take away). Gaps in the middle? Guess the 2nd number, test it, adjust.

a Chain (a): work backwards from 13 and 21. **b** Chain (b): guess the 2nd number, then check.

Investigation and Looking back

Investigation

Make the target. ***Find all the ways. Make your own and swap.*** **Use ALL four numbers 2, 3, 4 and 5, once each. You may use +, –, ×, ÷ and brackets. One is done for you: 2 + 3 + 4 – 5 = 4.**

> **Easy first step.** Use only + and –. With 2, 3, 4 and 5 (each once) can you make 6? 8? 10? Write a number sentence for each. Then carry on with the parts below, as far as you like.

a Make as many of the target numbers from 1 to 12 as you can. Write a number sentence for each target.

b Which target was hardest? Is there a target from 1 to 12 that you think cannot be made? Say why.

c Swap: write a target from 1 to 20 for a partner. Check their number sentence by working it out.

Looking back

Looking back 2 minutes at the end. You can write, tick or pass. Nobody marks this. It is not shown to the group. Your tutor may look at it on camera if you hold it up.

Something I did well _____

The hardest part was _____

What helped me ☐ drawing it ☐ a list or table ☐ a smaller number ☐ my partner ☐ reading it again ☐ I asked for help ☐ other: _____

☐ **I am still stuck** (my tutor will check in with me) ☐ **I pass today** (that is fine)

Story corner: Katherine Johnson USA (West Virginia and Virginia), 1918–2020

Katherine Johnson was a NASA mathematician who worked out flight paths for early space flights. Accounts say astronauts trusted her to check calculations a second way.

Try it (optional): Check that 6 × 48 = 288 using a different method from your first one. Hint: 48 is close to 50.

Think harder

Draw a model, make a list or make a table. Then write how you know. Choose these any time, or take them home.

No.1

There are 2 340 red and blue beads. There are 4 times as many blue beads as red beads. How many red beads are there?

No.2

Put ONE pair of brackets in 3 + 5 × 6 – 2 to make the answer 46.

No.3

Write +, –, × or ÷ in each gap to make this true: 12 ___ 4 ___ 5 = 15

No.4

Think of a number. Multiply it by 5. Add 35. Divide by 5. Take away the number you started with. The answer is always 7. Explain why it always works. A bar model can help.

Go further

No.5

Put + or – in each gap so that the answer is 7: 1 ___ 2 ___ 3 ___ 4 ___ 5 ___ 6 = 7. Find ALL the ways.

No.6

Mei spent half of her money on a book. Then she spent $6 on lunch. Then she spent half of what was left on a gift. She had $9 left. How much money did she have at first?

No.7

The number trick: think of a number, multiply by 5, add 35, divide by 5, take away your number. It always ends with 7. (a) What must you change so that it always ends with 10? (b) Make up your own number trick with four steps that always ends with 3. Test it with two numbers and explain why it works.

FRACTION FRONTIER

Name _____ Group _____ Date _____

Page 1: Start here. *After question 3 you may choose: carry on, or try Think harder.*

Improper fraction: 1 or more, like $\frac{7}{4}$. **Mixed number:** a whole number and a fraction, like $1\frac{3}{4}$.

No.1

Each bar is one whole. Draw more bars if you need them.



a Fifths shaded: _____ Improper fraction: _____ Mixed number: _____

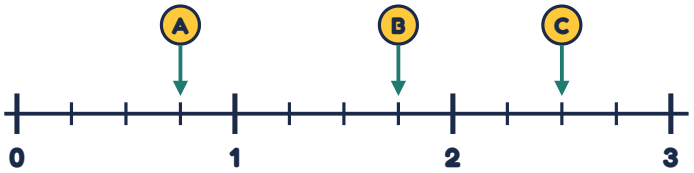
b Write $\frac{19}{4}$ as a mixed number. _____ c Write $3\frac{2}{5}$ as an improper fraction.

_____

d How many eighths are there in 5 wholes? _____

No.2

This number line is marked in quarters.



a Write the number at each arrow. A = _____ B = _____ C = _____

b Mark $\frac{9}{4}$ on the number line with an X.

No.3

Fraction of a set. There are 24 children in a class. $\frac{3}{8}$ of them wear glasses. Use the bar model.



a How many wear glasses? _____ b How many do NOT wear glasses? _____

Page 2 is on your own. Aim for Q4 and Q5 first. Q6 is a choice for the last few minutes.

No.4

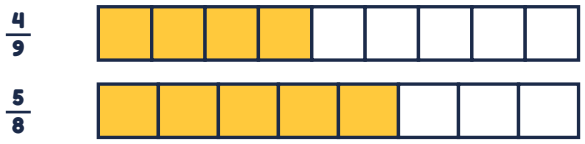
Compare. Write > or < in each box. Your fraction wall or a drawing helps. In (c), write how you know.

a $\frac{3}{4}$ ☐ $\frac{5}{8}$

b $\frac{2}{3}$ ☐ $\frac{7}{12}$

c $\frac{4}{9}$ ☐ $\frac{5}{8}$ The two bars below are the same length. Write how you know:

_____



d $\frac{3}{5}$ ☐ $\frac{3}{7}$

e Put in order, smallest first: $\frac{2}{3}$, $\frac{5}{6}$, $\frac{7}{12}$ _____, _____, _____

No.5

$\frac{3}{5}$ of the sweets in a jar are red. There are 27 red sweets.

a Draw a bar model. How many sweets are in the jar?

Answer: _____

b How many sweets are NOT red?

Answer: _____

No.6

Find ALL the fractions with denominator 12 that are bigger than $\frac{1}{3}$ but smaller than $\frac{3}{4}$. Hint: draw a strip of 12 equal boxes and shade $\frac{1}{3}$ of it, then $\frac{3}{4}$.

.....

.....

How many? _____

Investigation and Looking back

Investigation

Sharing 24 stickers. ***What if we change …? What stays the same?*** **Mei has 24 stickers. She gives 1/4 of them to Ali. That is 6 stickers. What if she gives a different fraction?**

> **Easy first step.** Draw 24 dots. Circle them in 2 equal groups: how many dots are in one group? That is 1/2 of 24. Now try 3 equal groups. Then carry on with the parts below, as far as you like.

a How many stickers does she give if she gives 1/2? _____ 1/3? _____ 1/6? _____ 1/8? _____ 1/5? _____ What do you notice about 1/5?

b Which of 1/2, 1/3, 1/4, 1/5 … up to 1/12 of 24 stickers give a whole number of stickers? List ALL of them.

c Mei gets more stickers. What is the smallest number of stickers so that 1/2, 1/3, 1/4, 1/5 AND 1/6 of them are all whole numbers?

.....

.....

.....

.....

Looking back 2 minutes at the end. You can write, tick or pass. Nobody marks this. It is not shown to the group. Your tutor may look at it on camera if you hold it up.

Something I did well _____

The hardest part was _____

What helped me ☐ drawing it ☐ a list or table ☐ a smaller number ☐ my partner ☐ reading it again ☐ I asked for help ☐ other: _____

☐ **I am still stuck** (my tutor will check in with me) ☐ **I pass today** (that is fine)

Story corner: Mahavira India (Karnataka, possibly Mysore), about 800 - about 870 (he worked around 850)

Mahavira, an Indian maths writer from about 1,200 years ago, wrote a book that explains how to work with fractions. It also shows how to split a fraction into unit fractions.

Try it (optional): Show that 1/2 = 1/3 + 1/6 by making the denominators the same. Then split 3/4 into 1/2 plus a unit fraction.

Think harder

Draw a model, make a list or make a table. Then write how you know. Choose these any time, or take them home.

No.1

Ben ate 7/4 pizzas and Ravi ate 1 5/8 pizzas. Who ate more?

No.2

Siti has 40 beads. 3/8 of them are blue. How many beads are NOT blue?

No.3

Put these fractions in order, smallest first: 2/3, 1/2, 5/6, 7/12

No.4

Who am I? I am an improper fraction. My denominator is 6. I am between 2 and 3. My numerator is an odd number. I am closer to 3 than to 2. Make a list, then cross out the ones that do not fit.

Go further

No.5

Mei takes 1/2 of the sweets in a jar. Ali takes 1/3 of the sweets. There are 5 sweets left. (a) How many sweets were in the jar at first? (b) What if 7 sweets were left? (c) Make up your own puzzle with two other fractions taken from a jar and a number of sweets left, and solve it.

No.6

Half of a number is 3 more than a third of the number. What is the number?

No.7

Siti read 2/5 of her book on Monday. On Tuesday she read half of the pages that were left. She still has 45 pages to read. How many pages are in the book?

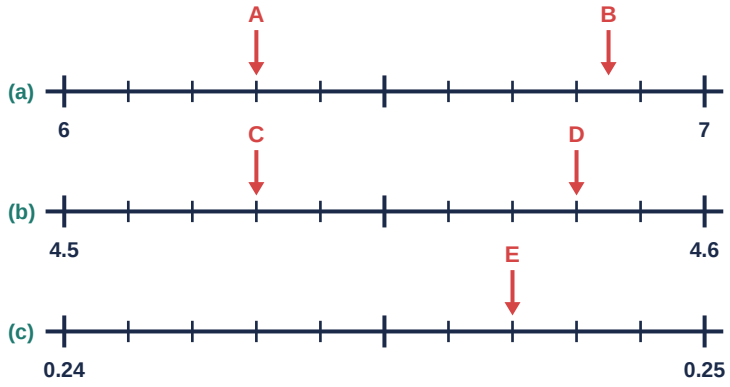
DECIMAL DETECTIVES

Name _____ Group _____ Date _____

Page 1: Start here. *After question 3 you may choose: carry on, or try Think harder.*

No.1

Zoom in! Write the number that each arrow points to. First work out what one small step is worth.



A = _____ B = _____ C = _____ D = _____ E = _____

f Which arrow, A to E, shows a number that rounds to 7? _____

No.2

Order and compare.

a Write in order, smallest first: 3.405, 3.45, 3.054, 3.5, 3.045

_____, _____, _____, _____, _____

b Ben says 3.405 is bigger than 3.45 because 405 is bigger than 45. Is he right? Explain.

.....

No.3

The rounding detective. A number has 2 decimal places. Rounded to 1 decimal place, it is 7.4. Use the number line: it is marked in hundredths.



a What is the smallest it can be? _____ The largest? _____

Page 2 is on your own. Aim for Q4 and Q5 first. Q6 is a choice for the last few minutes.

No.4

Use the digit cards 3, 6, 9 and a decimal point card once each. The point goes after the FIRST digit, like 3.69.

a Make all 6 numbers and list them in order, smallest first.

b Which is the largest number? _____

c Which of your numbers is closest to 7? _____

d Two of your numbers round to 4 (to the nearest whole number). Which two?
_____ _____

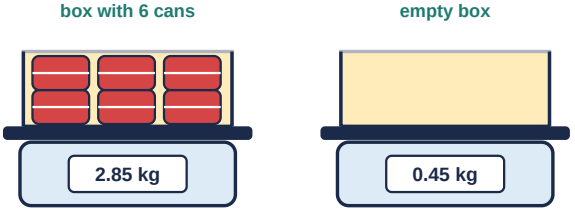
No.5

Mei and Siti have $9.40 altogether. Mei has $1.20 more than Siti. How much money does each girl have? Draw a bar model.

Mei $_____ Siti $_____

No.6

A box holds 6 cans that all have the same mass.



Step 1: the 6 cans alone. 2.85 – 0.45 = _____ kg. Step 2: share it between the 6 cans.



a What is the mass of one can? _____ kg

b Ali takes 2 cans out of the box. What is the mass of the box and cans now?
_____ kg

Investigation and Looking back

Investigation

Between two decimals. ***Is it always true? Test and explain.*** **Siti says: "There is no decimal between 0.35 and 0.36, because they are next to each other."**

> **Easy first step.** Write 0.35 as 0.350 and 0.36 as 0.360. Now write ONE decimal that sits between them. Then carry on with the parts below, as far as you like.

a Find a decimal between 0.35 and 0.36. Write it.

b Find as many decimals with 3 decimal places as you can between 0.35 and 0.36. Write them in order. How many are there?

c Is Siti right? Can you find a decimal between 0.35 and your smallest one? What do you think happens if you keep going?

.....

.....

.....

.....

Looking back 2 minutes at the end. You can write, tick or pass. Nobody marks this. It is not shown to the group. Your tutor may look at it on camera if you hold it up.

Something I did well _____

The hardest part was _____

What helped me ☐ drawing it ☐ a list or table ☐ a smaller number ☐ my partner ☐ reading it again ☐ I asked for help ☐ other: _____

☐ **I am still stuck** (my tutor will check in with me) ☐ **I pass today** (that is fine)

Story corner: Simon Stevin Flanders (born in Bruges, now Belgium), 1548–1620

Simon Stevin, from Bruges, wrote a small book in 1585 that explained decimals clearly. He showed how to split a whole into tenths, hundredths and thousandths.

Try it (optional): Write 3 + 4/10 + 5/100 as a decimal. Then add 2.8 to your answer.

Think harder

Draw a model, make a list or make a table. Then write how you know. Choose these any time, or take them home.

No.1

Find ALL the digits that the letter A can be: 3.A5 < 3.42 (A is one digit)

No.2

A number with 2 decimal places rounds to 10 when rounded to the nearest whole number. What is the smallest it can be? The largest?

No.3

What number is halfway between 2.6 and 2.7? What number is halfway between 2.6 and 2.61?

No.4

Mystery decimal. I have 3 decimal places and I am between 4 and 5. Rounded to 1 decimal place, I am 4.7. My hundredths digit is 2 more than my tenths digit. My digits add up to 25. Who am I?

Go further

No.5

Make up your own "closest to" puzzle. Choose four digit cards and a decimal point (the point goes between two digits). Pick a target number and find the number with 2 decimal places that is closest to it. Then ask: what if the target changes? Swap puzzles with a partner and check each other's answers.

No.6

Ali, Ben and Siti share $9.20. Ben gets $0.50 more than Ali. Siti gets $1.30 more than Ben. How much does each child get?

No.7

A jug full of water weighs 1.45 kg. When it is half full, it weighs 0.85 kg. What is the mass of the empty jug?

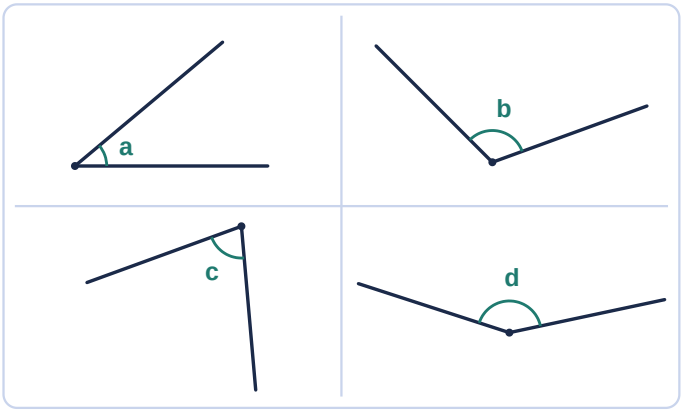
ANGLE TURNS

Name _____ Group _____ Date _____

Page 1: Start here. *After question 2 you may choose: carry on, or try Think harder.*

No.1

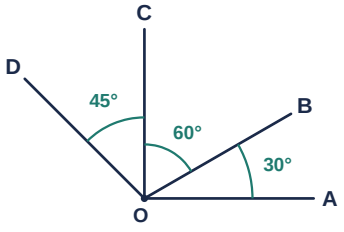
Estimate each angle first. Then measure it with your protractor. The arms are short: draw them longer with a ruler first. Tip: put the centre of the protractor on the corner, and count up from the 0 that lines up with one arm.



Angle	a	b	c	d
My estimate				
I measured				

No.2

Four rays OA, OB, OC and OD come out of point O. ∠AOB means the angle at O between OA and OB.



a How many angles can you find altogether? _____

b Name every angle and find its size. Work in order so you do not miss one.

c Which angle is a right angle? _____

Page 2 is on your own. Aim for Q3 and Q4 first. Q5 and Q6 are choices for the last few minutes.

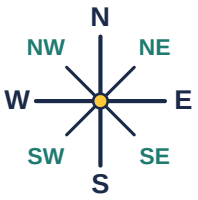
No.3

Clock angles. A clock face has 12 equal spaces in 360°. Find the smaller angle.

a At 4 o'clock: _____ **b** At 7 o'clock: _____

No.4

Compass turns. One step from a direction to the next one is 45°.

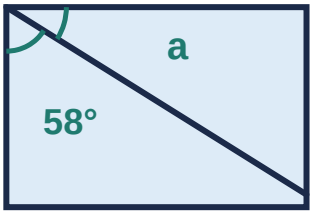


a Ali faces W and turns 90° clockwise. Now he faces: _____

b Mei faces SW and turns 135° anticlockwise. Now she faces: _____

No.5

The corner of a rectangle is 90°. Find the angle marked a.



∠a = _____

No.6

A robot starts facing N.

It turns 90° clockwise, then 45° anticlockwise, then 180°. Now it faces: _____

Investigation and Looking back

Investigation

Splitting a corner. ***Find all the ways.*** **A square corner is 90°. Mei cuts it into three angles. Every angle is a whole multiple of 10°: 10°, 20°, 30° and so on. One way is 10°, 20° and 60°. (30°, 50°, 10° and 10°, 30°, 50° count as the same set.)**

> **Easy first step.** A square corner is 90°. Draw it and split it into three EQUAL angles. How big is each one? Then carry on with the parts below, as far as you like.

a Find as many different sets of three angles as you can.

b How do you know you have found them all? Write how you checked.

c What if the corner is a straight line (180°) cut into three angles that are multiples of 30°? Find all the sets. What stays the same?

Looking back 2 minutes at the end. You can write, tick or pass. Nobody marks this. It is not shown to the group. Your tutor may look at it on camera if you hold it up.

Something I did well _____

The hardest part was _____

What helped me ☐ drawing it ☐ a list or table ☐ a smaller number ☐ my partner ☐ reading it again ☐ I asked for help ☐ other: _____

☐ **I am still stuck** (my tutor will check in with me) ☐ **I pass today** (that is fine)

Story corner: Eratosthenes Cyrene (now Libya) and Alexandria, Egypt; ancient Greek world, about 276 BC – about 194 BC

Eratosthenes, a Greek scholar in Egypt over 2,000 years ago, used the angle of a shadow to work out how big the Earth is. His angle was one fiftieth of a whole turn.

Try it (optional): A whole turn is 360 degrees. If a shadow made an angle of 10 degrees, how many angles that size fit all the way round?

Think harder

Draw a model, make a list or make a table. Then write how you know. Choose these any time, or take them home.

No.1

Siti faces SE. She turns 225° clockwise. Which direction does she face now?

No.2

What is the angle between the clock hands at 10 o'clock? At 2 o'clock? What do you notice?

No.3

How many degrees does the minute hand turn from 7:15 to 7:50?

No.4

A robot starts facing N. Ben wants it to turn from N to W using two DIFFERENT cards from: 45° clockwise, 90° clockwise, 135° anticlockwise, 180°. Find ALL the pairs that work.

Go further

No.5

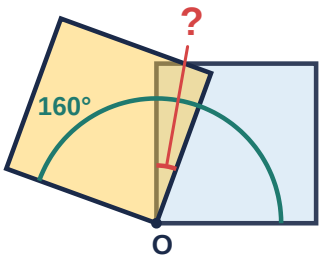
Make up your own clock-angle puzzle. Choose a time that is NOT an o'clock time (half past is easiest), work out the angle between the hands, and write your puzzle as "What is the angle at ...?". Draw the clock to check your answer. Then find a second time with the same angle.

No.6

The corner of a square is cut into 3 angles. The middle angle is twice the smallest. The biggest is 3 times the smallest. Find all three angles.

No.7

Two squares share one corner O and overlap, like the picture. The angle covered by both squares together (from the outside side of one square to the outside side of the other) is 160°. What is the angle where the squares overlap?



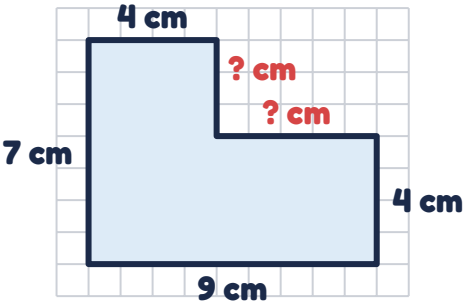
AREA ARCHITECTS II

Name _____ Group _____ Date _____

Page 1: Start here. *After question 3 you may choose: carry on, or try Think harder.*

No.1

This figure has only square corners. Each small square on the grid is 1 cm by 1 cm.



a Use the opposite sides to find the two missing lengths. Write them on the figure.

b Find the area in TWO ways.

Split it into 2 rectangles:

_____

Big rectangle take away the missing piece:

_____

c Find the perimeter. Perimeter: _____

No.2

Rectangles with whole-cm sides have an area of 24 cm². Find them all. Start with a length of 1 cm.

Length (cm)					
Breadth (cm)					
Perimeter (cm)					

a Largest perimeter: ____ by ____ cm Smallest perimeter: ____ by ____ cm

No.3

Mei joins 6 square tiles, each 1 cm by 1 cm, edge to edge (whole sides touching).

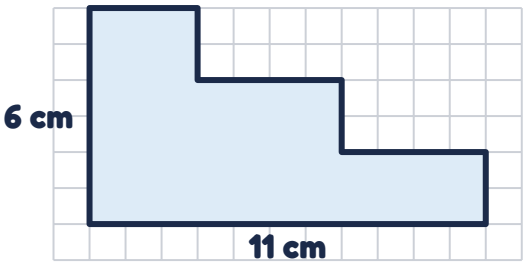
a What is the area of any shape she makes? _____

b Draw the shape with the SMALLEST perimeter and one with the LARGEST. Smallest: _____ Largest: _____

Page 2 is on your own. Aim for Q4 and Q5 first. Q6 is a choice for the last few minutes.

No.4

Ravi says, "I cannot find the perimeter of this staircase, because the steps are not labelled."



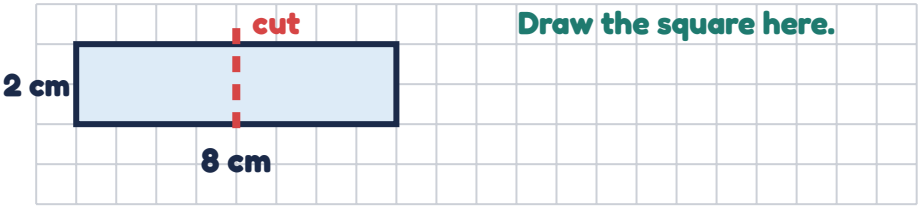
a Show Ravi how to find the perimeter WITHOUT measuring each step. Push the steps out: the shape then fits in a ____ cm by ____ cm rectangle.

____ + ____ + ____ + ____ = ____ cm.
Perimeter: _____

b Think of the staircase as 3 rows. The rows are 3 cm, 7 cm and 11 cm long, and each row is 2 cm high. Area: _____

No.5

An 8 cm by 2 cm rectangle is cut along the dashed line. The two pieces are put together to make a square.



a Draw the square on the grid.

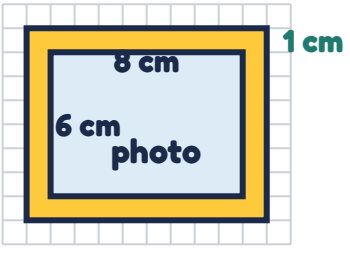
b Area: rectangle _____ cm², square _____ cm². Perimeter: rectangle _____ cm, square _____ cm.

c Ali says, "The area stayed the same, so the perimeter stayed the same too." Is he right? Explain.

.....

No.6

A photo 8 cm long and 6 cm wide is put in a frame. The frame is 1 cm wide all round.



a How long and how wide is the whole thing (photo and frame)? _____ cm by _____ cm

b Find the area of the frame only. Area of frame: _____

c Mei found the frame like this: 8 + 8 + 6 + 6 = 28 cm². What did she forget? _____

Investigation and Looking back

Investigation

Bigger area, bigger perimeter? ***Is it always true? Test and explain.*** **Ali says: "A rectangle with a bigger area always has a bigger perimeter."**

> **Easy first step.** Draw a 4 cm by 4 cm square on the Square grid. Write its area and its perimeter. Then carry on with the parts below, as far as you like.

a Test Ali's idea. Find the area and perimeter of a 1 cm by 11 cm rectangle, and of a 4 cm by 4 cm square. Which has the bigger area? Which has the bigger perimeter?

b List ALL the rectangles with whole-cm sides and a perimeter of 24 cm. Write the area of each.

c Is Ali right? Which of your rectangles has the biggest area? What is special about its shape?

Looking back

2 minutes at the end. You can write, tick or pass. Nobody marks this. It is not shown to the group. Your tutor may look at it on camera if you hold it up.

Something I did well _____

The hardest part was _____

What helped me ☐ drawing it ☐ a list or table ☐ a smaller number ☐ my partner ☐ reading it again ☐ I asked for help ☐ other: _____

☐ **I am still stuck** (my tutor will check in with me) ☐ **I pass today** (that is fine)

Story corner: Euclid Alexandria, Egypt (Greek-speaking world), about 325 BC – about 265 BC (dates are estimates)

Euclid taught maths in Alexandria about 2,300 years ago. His book the Elements shows that a slanted shape with the same base and height as a rectangle has the same area.

Try it (optional): A rectangle is 8 cm long and 5 cm high. Slide its top side along to make a slanted shape with the same base and height. What is the area of each? Which has the longer way round?

Think harder

Draw a model, make a list or make a table. Then write how you know. Choose these any time, or take them home.

No.1

List every rectangle with whole-cm sides and an area of 30 cm². Which one has the smallest perimeter?

No.2

A rectangle has whole-cm sides and a perimeter of 20 cm. List all the possible rectangles and their areas. Which has the largest area?

No.3

A 3 cm by 3 cm square is cut from each of the four corners of a 12 cm by 8 cm rectangle. Find the area and the perimeter of the shape that is left.

No.4

Three squares with sides 2 cm, 3 cm and 4 cm stand side by side in a row on the same line, smallest on the left. Find the area and the perimeter of the figure.

Go further

No.5

Make up your own fencing puzzle like Ravi's: choose a length of fence (a whole number of metres, more than 20) and say whether one side is a wall. Find the largest area the pen can have, and prove it by listing every whole-metre pen. Swap puzzles with a partner and check each other's lists.

No.6

A garden is 10 m long and 8 m wide. Two paths 1 m wide cross it: one goes along its length and one goes across its width. What area of the garden is NOT path?

No.7

A rectangle 9 cm long and 4 cm wide is cut into 2 pieces. The pieces fit together exactly to make a square. How long is a side of the square? Did the perimeter go up or down?

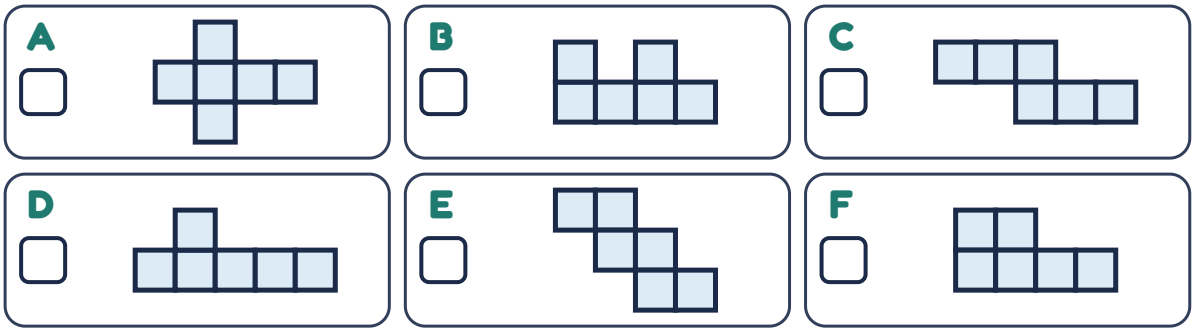
SYMMETRY AND NETS

Name _____ Group _____ Date _____

Page 1: Start here. *After question 3 you may choose: carry on, or try Think harder.*

No.1

Each shape is made of 6 squares. Which ones fold into a cube? Tick them.



a Shapes ticked: _____ Shape that does NOT fold: _____ . Explain why:

.....

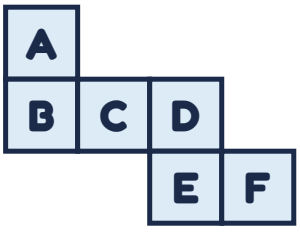
No.2

This net folds into a cube. Which letter is opposite ...

a A? _____

b B? _____

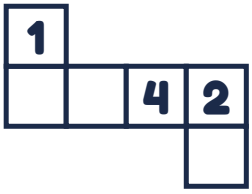
c C? _____



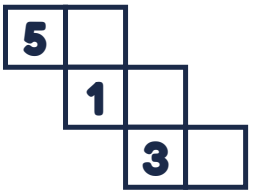
No.3

On a dice, the numbers on opposite faces add up to 7. Write the missing numbers on each dice net.

(a)



(b)



In net (b), how did you find the number opposite 1?

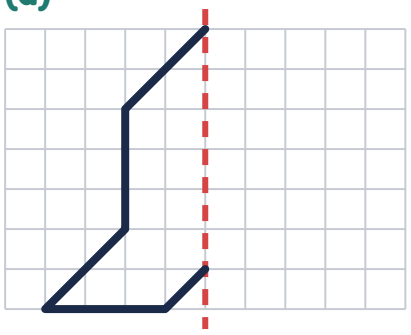
_____

Page 2 is on your own. Aim for Q4 and Q5 first. Q6 is a choice for the last few minutes.

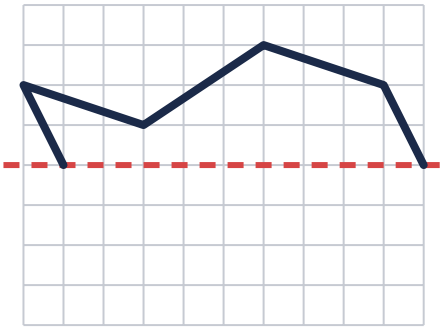
No.4

Complete each figure so that the dashed line is a line of symmetry.

(a)



(b)



Count the squares from the line to each corner. Put a dot the same number of squares on the other side, then join the dots.

No.5

Draw ALL the lines of symmetry on each shape. Then write how many.

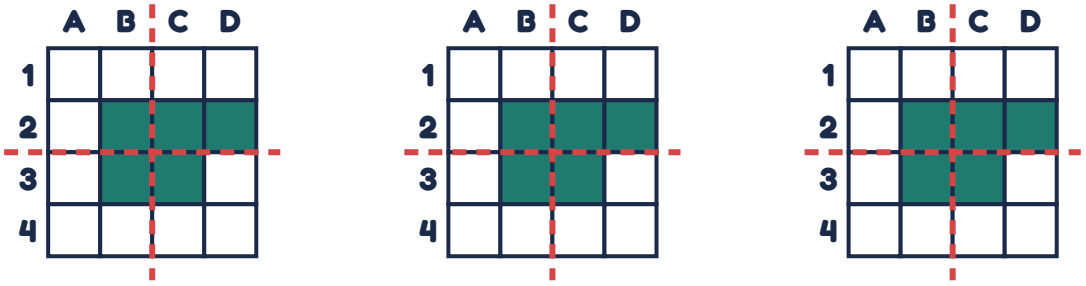
(a) **(b)** **(c)** **(d)**



(a) _____ (b) _____ (c) _____ (d) _____

No.6

Shade ONE more square so that a dashed line becomes a line of symmetry of the shaded squares. Find ALL the ways. Use a new grid for each.



a The squares I shaded (like A1): _____ How many ways? _____

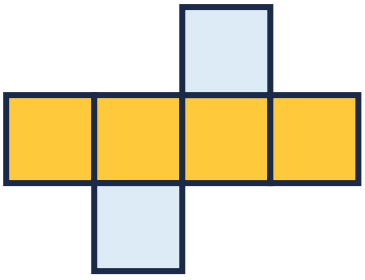
b Choose a square that does NOT work. Where would its partner have to be?

.....

Investigation and Looking back

Investigation

A row of four. ***Is it always true? Test and explain.*** **Here is a net: a row of 4 squares with one square above it and one square below it. Mei says: "A row of 4 squares with ONE square above it and ONE square below it always folds into a cube, wherever the two squares are."**



A row of 4 squares,
one square above it and
one square below it.

> **Easy first step.** Cut 6 squares from the Square grid page. Make the net in the picture (a row of 4 squares with one square above it and one below). Fold it. Does it make a cube? Then carry on with the parts below, as far as you like.

a Test Mei's idea. On squared paper (use the Square grid at the back), draw 4 different nets like this by moving the top square and the bottom square. Fold each one in your head. Do they all fold into a cube?

b Explain why it works. Use the words wall, lid and base.

c What if BOTH extra squares are above the row? Try two. Do they ever fold into a cube? Why or why not?

Looking back 2 minutes at the end. You can write, tick or pass. Nobody marks this. It is not shown to the group. Your tutor may look at it on camera if you hold it up.

Something I did well _____

The hardest part was _____

What helped me ☐ drawing it ☐ a list or table ☐ a smaller number ☐ my partner ☐ reading it again ☐ I asked for help ☐ other: _____

☐ **I am still stuck** (my tutor will check in with me) ☐ **I pass today** (that is fine)

Think harder

Draw a model, make a list or make a table. Then write how you know. Choose these any time, or take them home.

No.1

A cuboid is 3 cm long, 2 cm wide and 1 cm high. Its net is drawn on 1 cm squared paper. How many squares does the net cover?

No.2

Ben puts a dice on the table. The top face shows 3 and the front face shows 5. What numbers are on the bottom face and on the back face?

No.3

A dice is on a table. Mei walks round it and can see the top and the four side faces. The numbers she can see add up to 19. What number is on the bottom face?

No.4

There are 5 different shapes made of 4 squares joined edge to edge (a line, a square, an L, a T and a zigzag). Which one has exactly 1 line of symmetry?

Story corner: Emmy Noether Germany (born in Erlangen), 1882-1935

Emmy Noether, born in Germany in 1882, kept studying maths even when women were not welcome at university. She showed that symmetry is useful as well as beautiful.

Try it (optional): How many lines of symmetry does a square have? How many does a rectangle that is not a square have?

Go further

No.5

Design your own dice. Draw a net of a cube that is not one of the nets in this lesson, and write the numbers 1 to 6 on it so that opposite faces add up to 7. Name each pair of opposite faces to prove it works. Then rub out two numbers and challenge a partner to put them back.

No.6

In how many ways can you shade exactly 2 squares of a 3 by 3 grid so that the up-down line through the middle is a line of symmetry?

No.7

Shade exactly 5 squares of a 3 by 3 grid so that the pattern has 4 lines of symmetry. There are 2 ways. Find both.

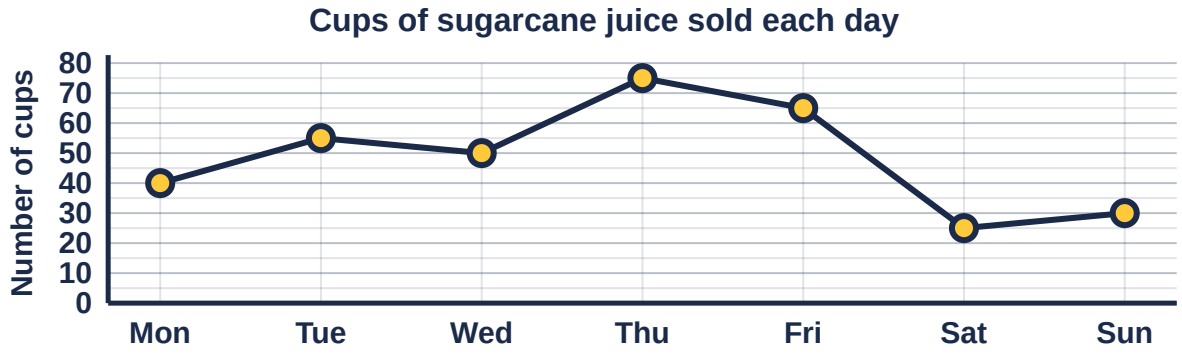
DATA DETECTIVES

Name _____ Group _____ Date _____

Page 1: Start here. *After question 2 you may choose: carry on, or try Think harder.*

No.1

A sugarcane juice stall is next to an office building. The line graph shows how many cups it sold each day for one week. Read the scale first!



a Cups sold on Thursday? _____ On Saturday and Sunday together? _____

b Biggest increase: from _____ to _____, up by _____ cups

c Biggest decrease: from _____ to _____, down by _____ cups

d How can you see (c) on the graph without working out every change? Why might sales drop on those days?

No.2

The table shows how many books three classes borrowed from the library. Fill in the missing numbers.

Class	Week 1	Week 2	Total
4A	38		85
4B		29	70
4C	45	52	
Total			

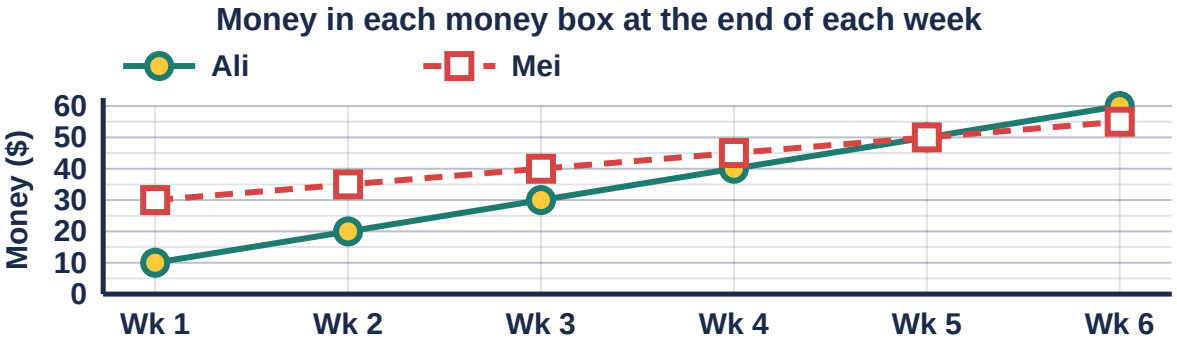
a Which class borrowed more in Week 1 than in Week 2? _____

b In Week 3, each class borrowed 6 more books than in Week 2. Week 3 total? _____

Page 2 is on your own. Aim for Q3 and Q4 first. Q5 is a choice for the last few minutes.

No.3

The graph shows how much money Ali and Mei had in their money boxes at the end of each week.



a Mei's money in Week 3? $_____ Ali saves each week? $_____

b Same amount in Week _____ Ali first has more in Week _____

c They keep saving the same way. In Week 8, Ali has $_____ more than Mei.

No.4

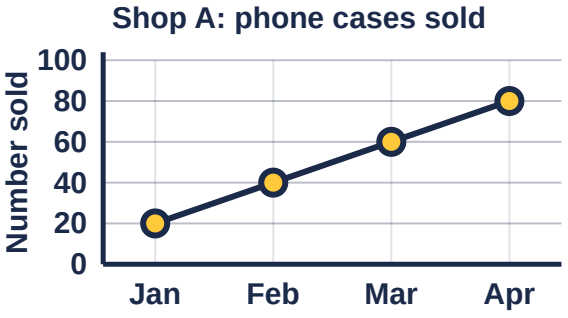
Siti reads more pages each day, always the same number more. Fill in the table.

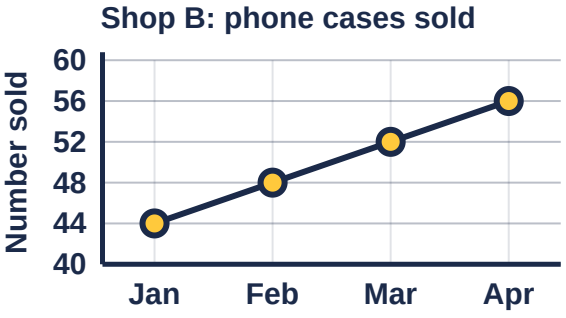
Day	1	2	3	4	5
Pages		14			26

a How many more each day? _____ Pages in the 5 days? _____

No.5

Two shops drew line graphs of the phone cases they sold. Ravi says, "Both shops' sales went up by the same amount, because the lines look the same."





a January to April: Shop A went up by _____. Shop B went up by _____.

b Is Ravi right? Explain why the two graphs look the same.

Investigation and Looking back

Investigation

Sales have doubled! ***What if we change ...? What stays the same?*** **A shop sold 50 phone cases in May and 100 in June. A newspaper says: "Sales have doubled!" (That is true: 100 is twice 50.) Another newspaper reports the same two numbers, but its headline sounds much smaller.**

> **Easy first step.** On the Graph grid, write 0, 10, 20 ... up the side. Plot the points (May, 50) and (June, 100) and join them. Then carry on with the parts below, as far as you like.

a On the first Graph grid, draw a line graph of May and June. Go up in 10s from 0 to 100. Is the line steep or gentle?

b Write TWO headlines about May and June that are both TRUE: one that sounds big, like "Sales have doubled!", and one that sounds small. Hint: how many MORE cases were sold in June?

c Which headline would you trust more? What would you ask the newspaper to add, so that readers are not tricked?

Looking back

2 minutes at the end. You can write, tick or pass. Nobody marks this. It is not shown to the group. Your tutor may look at it on camera if you hold it up.

Something I did well _____

The hardest part was _____

What helped me ☐ drawing it ☐ a list or table ☐ a smaller number ☐ my partner ☐ reading it again ☐ I asked for help ☐ other: _____

☐ **I am still stuck** (my tutor will check in with me) ☐ **I pass today** (that is fine)

Story corner: Florence Nightingale

Britain (born in Florence, Italy), 1820-1910

Florence Nightingale was a nurse who also loved numbers. She drew charts so that people could see the facts about hospitals at a glance.

Try it (optional): Temperatures: 9 am 26 degrees, noon 31 degrees, 3 pm 33 degrees, 6 pm 29 degrees. Between which two times did it rise the most, and by how much?

Think harder

Draw a model, make a list or make a table. Then write how you know. Choose these any time, or take them home.

No.1

A table shows that Class 4A has 28 boys and 6 more girls than boys. Class 4B has 63 pupils, with 5 more girls than boys. Which class has more girls?

No.2

On a line graph each line on the scale is worth 20. A point is 3 lines above the 40 line. What number does it show?

No.3

A shop sold 300 kites in 4 days. Day 1: 65, Day 2: 80, Day 4: 90. How many kites were sold on Day 3?

No.4

Ali's height: age 7, 120 cm; age 8, 127 cm; age 9, 131 cm; age 10, 137 cm. In which year did he grow the most? How much did he grow altogether from age 7 to age 10?

Go further

No.5

Four teams scored 100 points altogether. Team A scored twice as many as Team B. Team C scored 10 more than Team B. Team D scored 22. How many points did each team score?

No.6

Tank A has 80 litres of water and loses 6 litres every hour. Tank B has 20 litres and gains 9 litres every hour. On a line graph the two lines cross. After how many hours do the tanks hold the same amount? How much is in each?

No.7

Four numbers: 52, 54, 56, 58. Choose where the scale starts and how much each line is worth, so that a line graph of these numbers looks (a) like a huge increase and (b) like hardly any increase. Write both scales and a sentence a shop owner might say with each graph. Which graph is fair? Why?

CHALLENGE DAY

Name _____ Group _____ Date _____

Page 1: Start here. *After question 3 you may choose: carry on, or try Think harder.*

No.1

Guess and check. Make a table of your tries.

a Two consecutive numbers (next-door numbers, like 7 and 8) multiply to make 552. What are they?

Answer: _____

b Mei buys some $4 pens and some $7 notebooks for exactly $50. Find ALL the ways.

Notebooks	1	2	3	4	5	6	7
Money left for pens	$43						
Pens?							

Ways: _____

No.2

A farm has chickens and goats. There are 18 heads and 50 legs altogether.

a Suppose all 18 are chickens. Legs: _____ Legs short: _____

b Each swap to a goat adds _____ legs. So _____ goats and _____ chickens.

c A car park has cars and motorcycles: 24 vehicles, 70 wheels. How many cars?

Answer: _____

No.3

Work backwards. Start at the end and undo each step, last step first.

a Ravi gave away half of his marbles. Then he won 9 marbles and lost 5. Now he has 30. How many did he have at first?

Answer: _____

b I think of a number. I multiply it by 3, take away 18, then halve the answer. I get 21. What is my number?

Answer: _____

Page 2 is on your own. Aim for Q4 and Q5 first. Q6 is a choice for the last few minutes.

No.4

Ali, Ben, Mei and Siti each ordered a different dish: chicken rice, laksa, prata or nasi lemak. (Chicken rice and nasi lemak are the rice dishes.) Use the grid: ✓ for yes, X for no.

1 Ali and Siti both ordered rice dishes.

2 Ali did not order chicken rice.

3 Ben did not order prata.

	Chicken rice	Laksa	Prata	Nasi lemak
Ali				
Ben				
Mei				
Siti				

a Ali: _____ Ben: _____ Mei: _____ Siti: _____

No.5

Draw the queue.

a Mei is 7th from the front and 12th from the back of a queue at the MRT gantry. How many people are in the queue?

Answer: _____

b 20 people are in a queue. Ben is 6th from the front. What is his place from the back?

_____

No.6

Ben and his father are 50 years old altogether. In 5 years, his father will be 3 times as old as Ben. How old is Ben now? Draw bars.

Answer: _____

Investigation and Looking back

Investigation

Ring toss scores. ***Find all the ways.*** **In a ring-toss game, a small ring scores 3 points and a big ring scores 5 points. You can throw as many rings of each kind as you like. Which total scores are possible?**

> **Easy first step.** List the scores you can get with ONE ring: 3 or 5. Then with TWO rings: 3 + 3, 3 + 5 and 5 + 5. Write each total. Then carry on with the parts below, as far as you like.

a List the scores from 1 to 20 that you CAN get, and show how (for example 8 = 3 + 5). Circle the scores you can NOT get.

b What do you notice about the scores you cannot get? After which score can you get every number?

c What if the rings score 4 and 6? Which scores from 1 to 20 are impossible? Explain why.

Looking back 2 minutes at the end. You can write, tick or pass. Nobody marks this. It is not shown to the group. Your tutor may look at it on camera if you hold it up.

Something I did well _____

The hardest part was _____

What helped me ☐ drawing it ☐ a list or table ☐ a smaller number ☐ my partner ☐ reading it again ☐ I asked for help ☐ other: _____

☐ **I am still stuck** (my tutor will check in with me) ☐ **I pass today** (that is fine)

Story corner: Terence Tao Australia (born in Adelaide); now also works in the USA, born 1975

Terence Tao, from Adelaide, Australia, entered a maths Olympiad at age ten and won a bronze medal. He tried again each year and won silver, then gold.

Try it (optional): Use guess and check: a farm has hens and cows. Together they have 10 heads and 28 legs. How many hens and how many cows?

Think harder

Draw a model, make a list or make a table. Then write how you know. Choose these any time, or take them home.

No.1

There are 20 bicycles and tricycles with 47 wheels altogether. How many tricycles are there?

No.2

Two next-door even numbers (like 6 and 8) multiply to make 624. What are they?

No.3

Ravi spent $15 on a book. Then he spent half of the money left on a bag. He had $18 left. How much money did he have at first?

No.4

Ali, Ben and Siti are 9, 10 and 11 years old. Ben is older than Siti. Siti is not the youngest. How old is each child?

Go further

No.5

A hawker centre has 15 tables. Some seat 4 people and some seat 6 people. There are 76 seats altogether. How many tables seat 6?

No.6

Mei gave Ali half of her stickers and 2 more. Then she gave Ben half of the stickers left and 2 more. She had 5 stickers left. How many did she have at first?

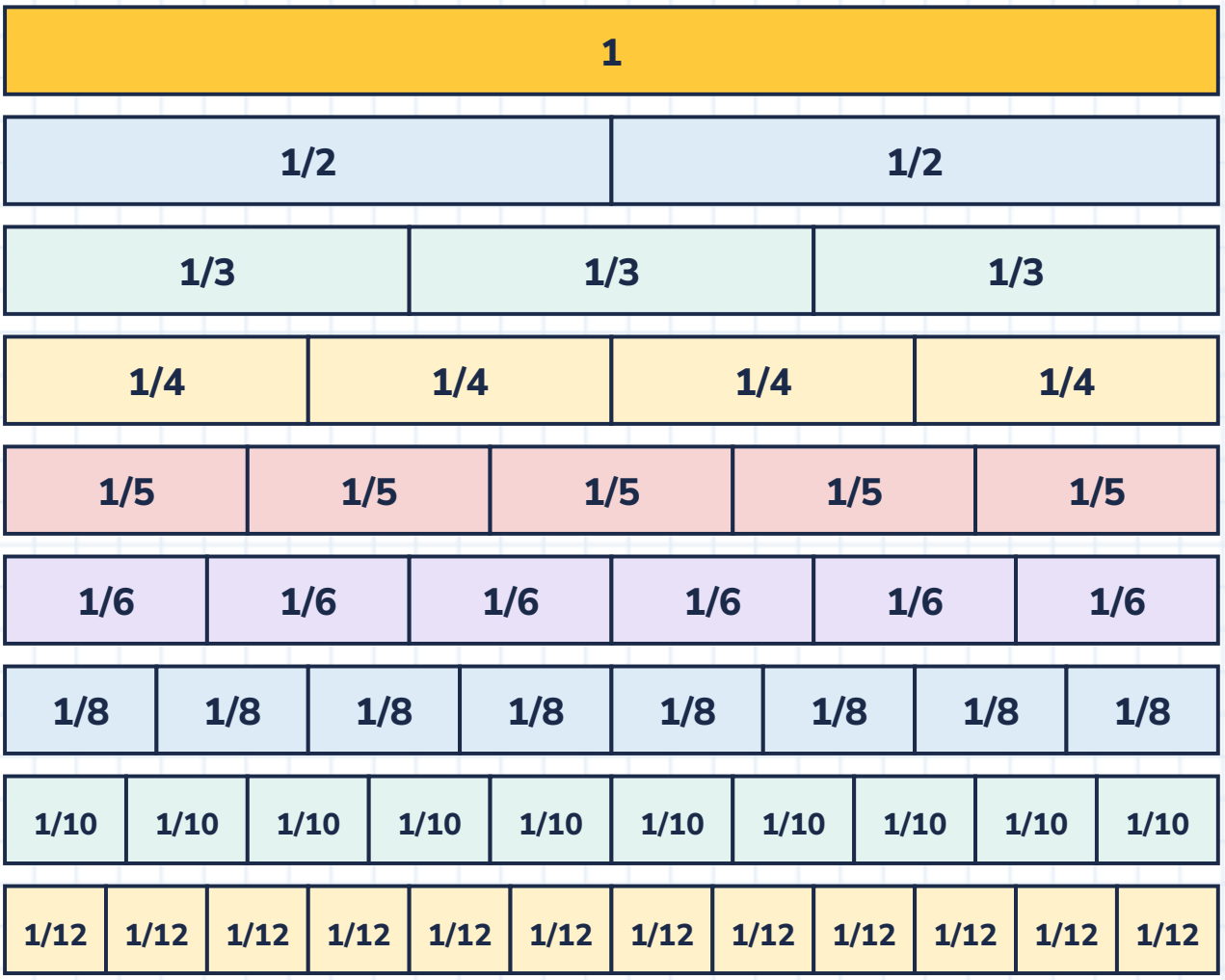
No.7

Make up your own heads-and-legs puzzle (chickens and goats, or bicycles and cars) that has exactly ONE answer. Check it by trying every number of one animal. Then explain why a farmer who counts 12 heads and 35 legs on a farm with only chickens and goats must have counted wrong.

Cut-outs

For the Level 4 program. Cut along the dashed lines and keep the pieces in one envelope.

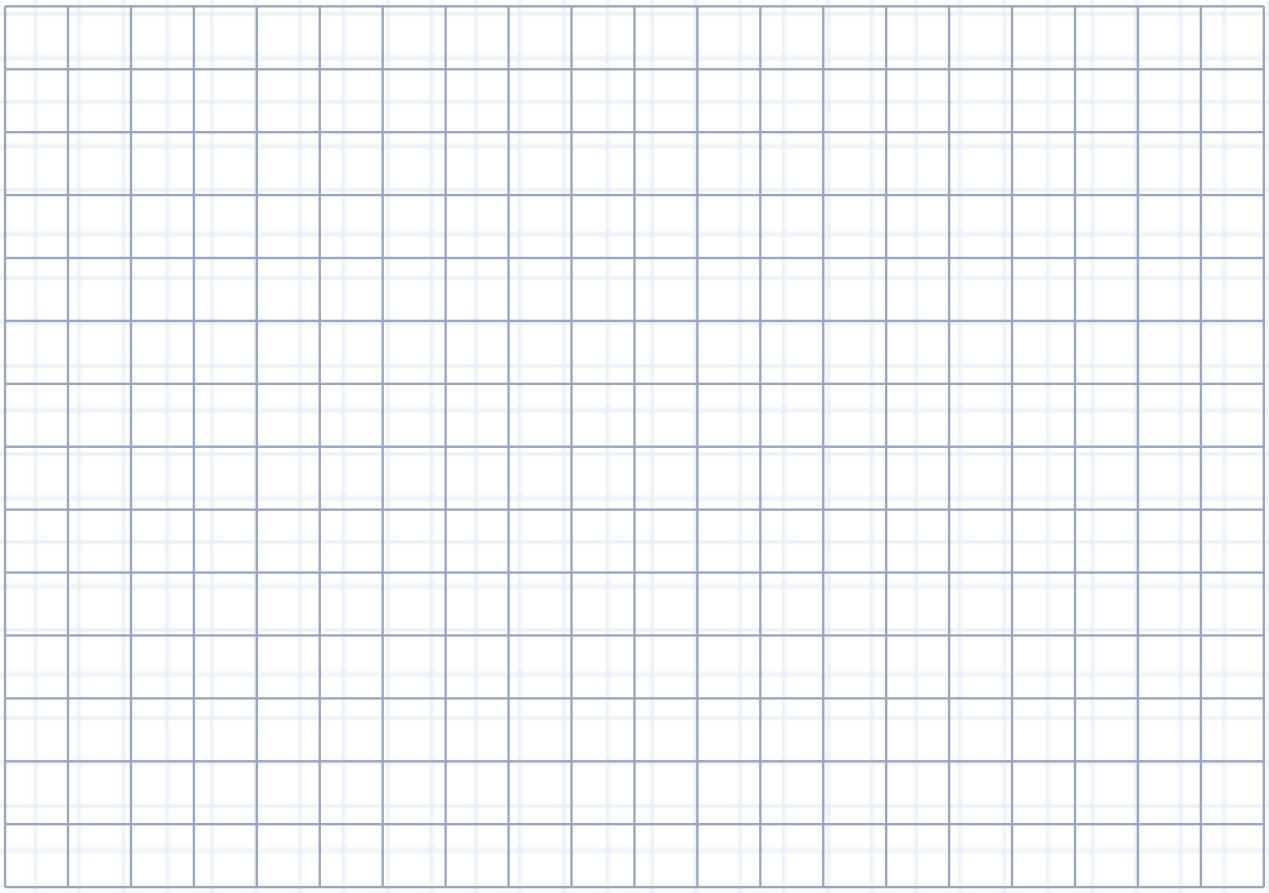
Fraction wall (Lesson 4)



Line the strips up under the whole. Which fractions are the same size?

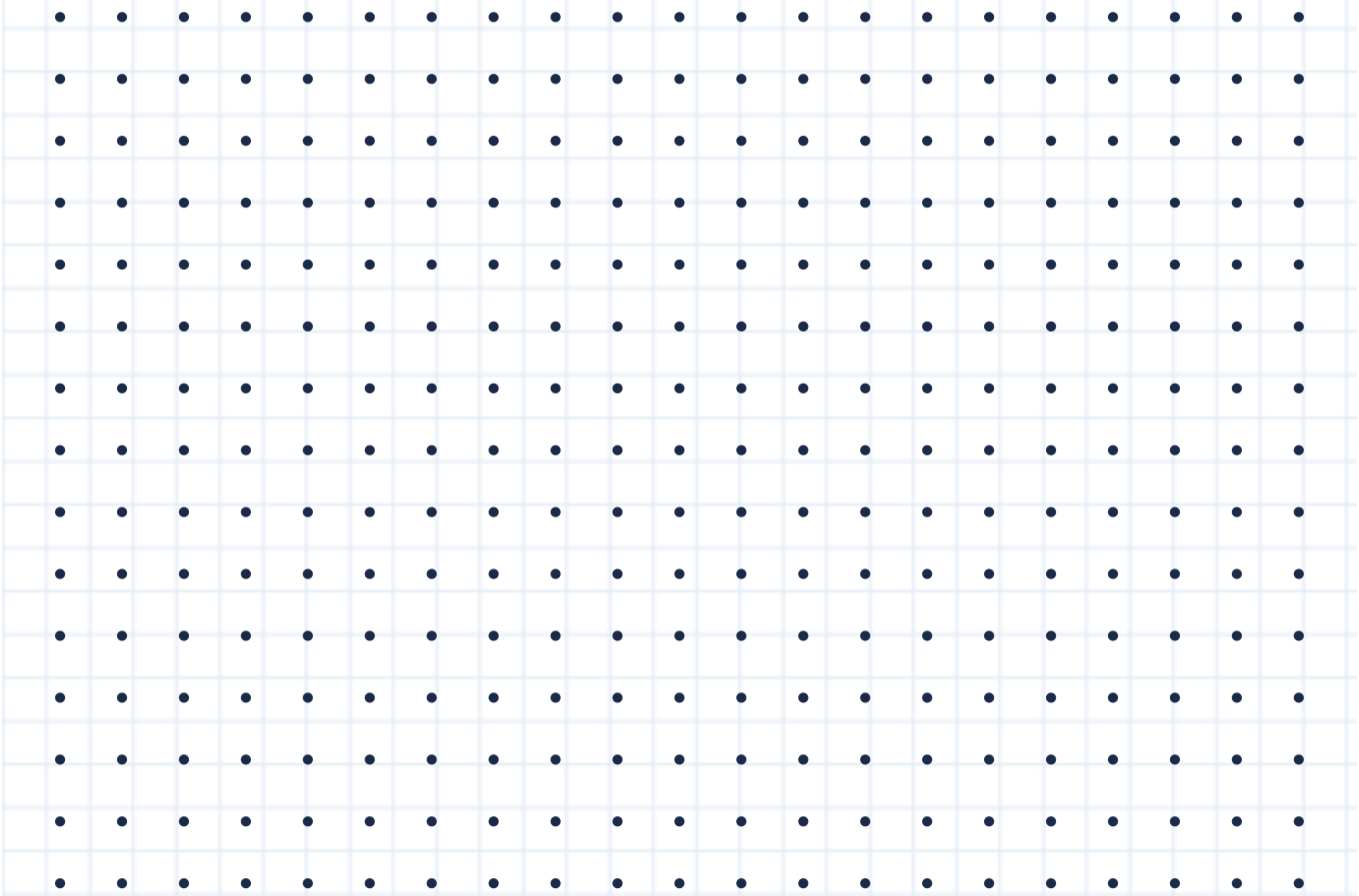
Square grid

For Lessons 7 and 8. Cut it into 1 cm squares if you need paper squares.



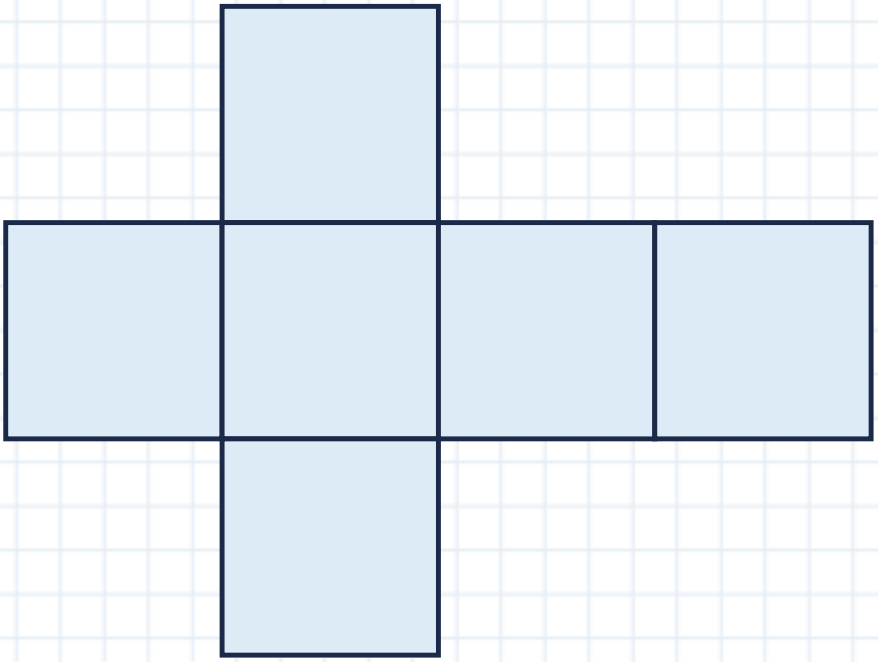
Dot grid

For Lesson 8 (symmetry).



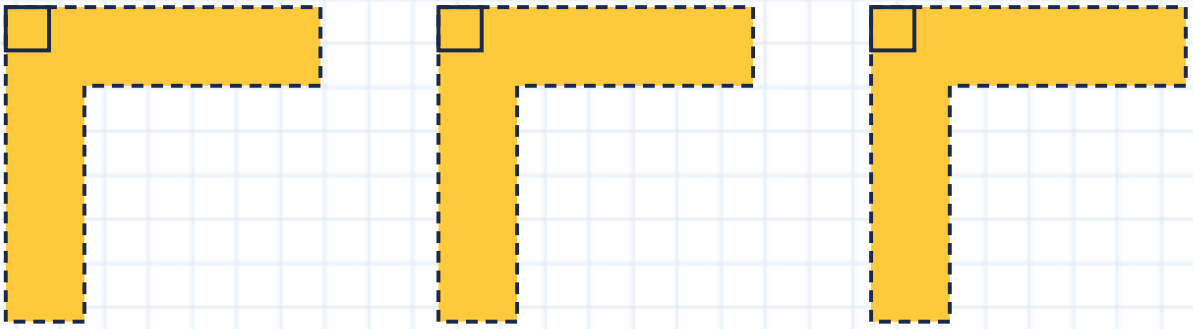
Cube net

For Lesson 8.



Cut round the outside, fold on the lines and tape: a cube. Try drawing other nets on the square grid.

Right-angle testers (Lesson 6)



Cut out a right-angle tester. Put its corner on an angle: bigger, smaller or exactly a right angle?

Digit, point, sign and bracket cards

For Lessons 2, 3, 5 and 10. Two sets of the digits 0 to 9, four decimal points, the four signs twice and two pairs of brackets.

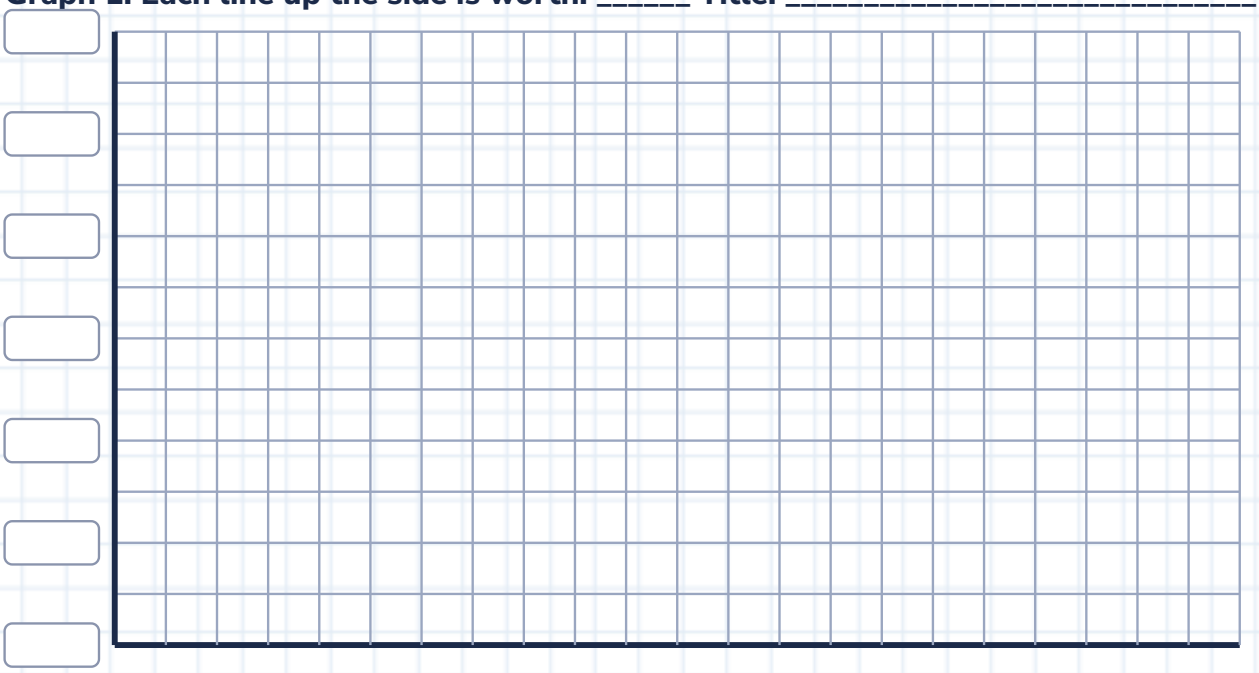
0	1	2	3	4	5
6	7	8	9	0	1
2	3	4	5	6	7
8	9	point .	point .	point .	point .
+	−	×	÷	+	−
×	÷	(	)	(	)

Cut along the dashed lines. Keep the cards in one envelope. Digits: Lessons 2, 3, 5 and 10. Decimal points: Les

Graph grids

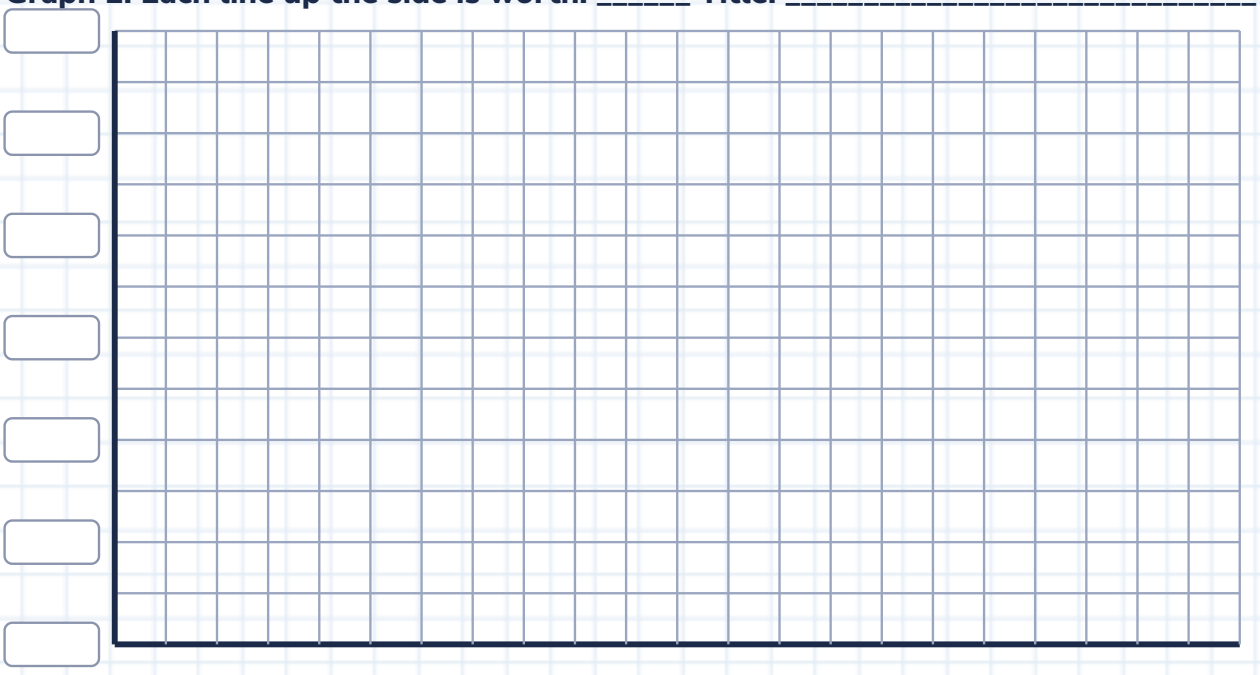
For Lesson 9. Fill in the scale before you plot.

Graph 1. Each line up the side is worth: _____ Title: _____



what is along the bottom: _____

Graph 2. Each line up the side is worth: _____ Title: _____

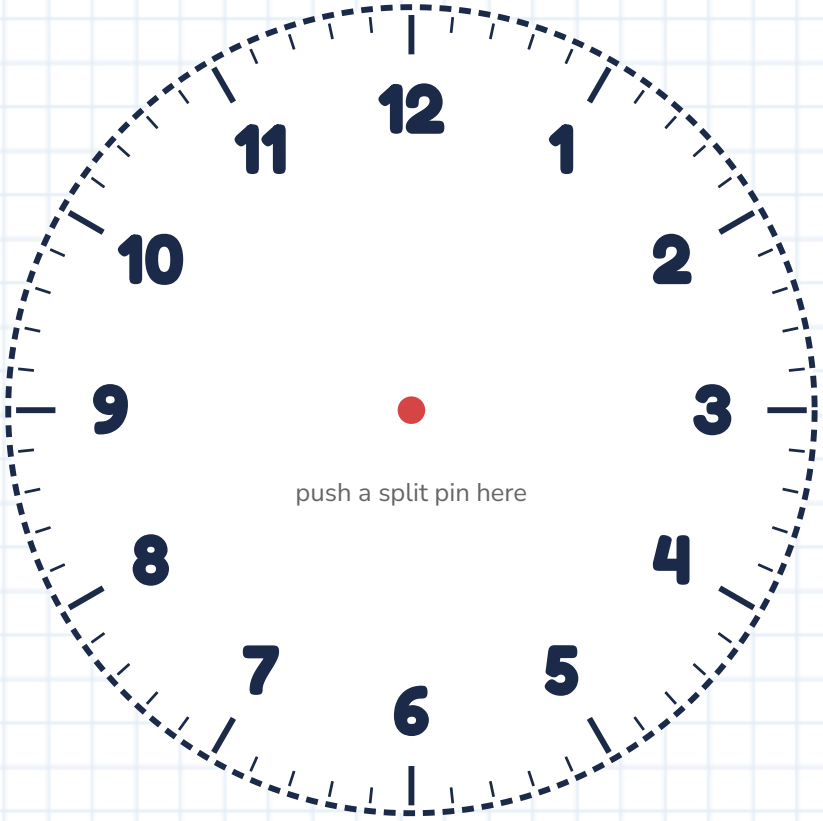


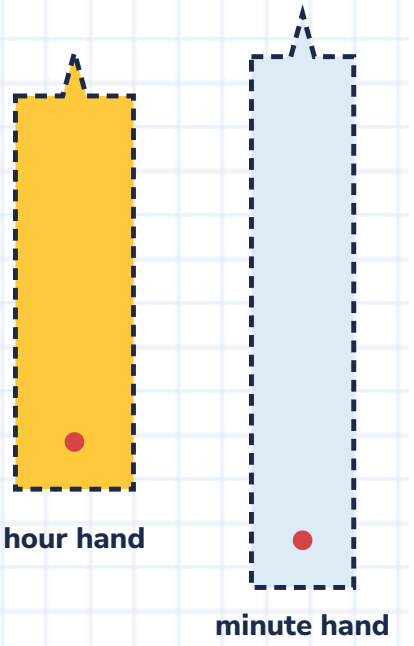
what is along the bottom: _____

Write the scale in the small boxes at the side first (for example 0, 10, 20 ...). Print extra copies if you need then

Paper clock, pizza circles and what to bring

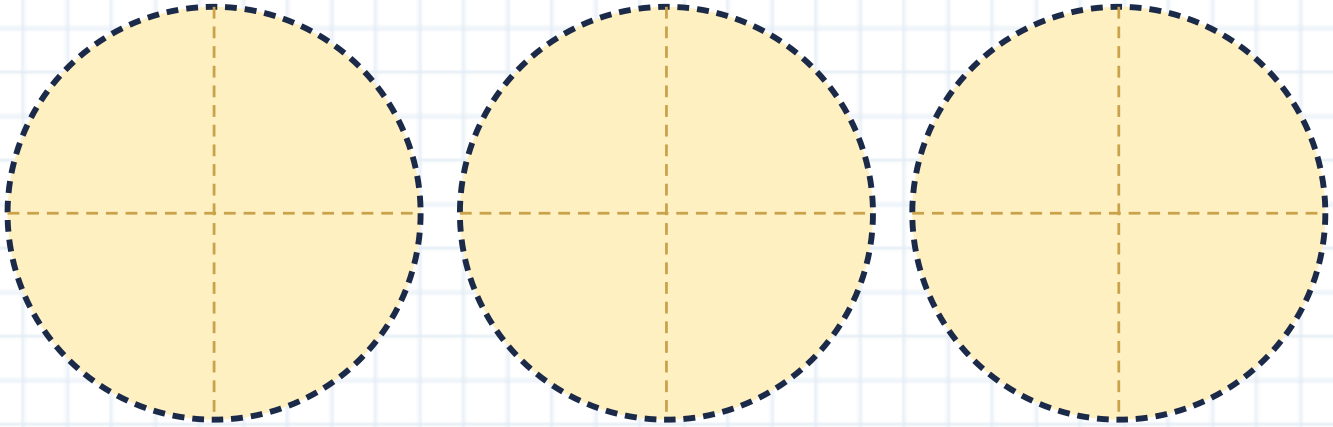
For Lessons 4 and 6.





Make a hole at each red dot, then jo

Pizza circles (Lesson 4): cut out each circle, then cut it into 4 equal slices (quarte



Faint lines show where to cut. Fold each circle in half twice to check the 4 slices are equal.

Bring from home (not printed here)

- A clear 180 degree protractor and a 30 cm ruler (Lesson 6). Ask your tutor if you do not have them
- A ruler and two coloured pencils (Lessons 7 and 9). Scissors and tape (Lessons 4, 6, 7 and 8).
- Squared paper, if you want extra: the Square grid page can be cut into 1 cm squares for Lessons 7

CERTIFICATE OF COMPLETION

Math Inquirers Holiday Program

Level 4 Holiday Program

This certifies that

.....

completed all 10 lessons, solved puzzles, spotted patterns and explained their thinking.

Notice it. Test it. Explain why.

Tutor: _____ Date: _____